"MATHEMATICAL MODELLING IN THE NATURAL SCIENCES"

LECTURE COURSE

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WHEN & WHERE

* EVERY 2 WEDNESDAY AND EVERY THURSDAY *
  * starting from 11.03.2026
  ⚠ follow UGO

START: 04.03.2026, END: 25 06.2026

* PLACE: SR 11 32

* TIME: WED 11:15 - 12:45
  THU 11:45 - 13:15

* OFFICE HOURS: THURSDAY AFTER CLASS + APPOINT.

FINAL EXAM

* THE EXAM IS ORAL

* BETWEEN 15.06.2026 AND 26.06.2026

* ~ 40 min



CONTENT

- ## INTRODUCTION AND MOTIVATIONS

  BASIC MODELS IN THE NATURAL SCIENCES,
  CHEMICAL REACTIONS: FROM THE REACTIONS TO THE EQUATIONS

- ## ODE MODELS IN BIOLOGY, ECOLOGY AND PHYSICS

  DERIVATION, ANALYSIS AND CLASSIFICATION OF THE STEADY STATES.
  APPLICATION IN EPIDEMIOLOGY AND POPULATION DYNAMICS

- ## PDE MODELS IN BIOLOGY, ECOLOGY AND PHYSICS

  DERIVATION, ANALYSIS OF THE STEADY STATES,
  TURING INSTABILITY AND PATTERN FORMATION.